

DNS in detail

What is DNS?

DNS stands for Domain name system. DNS provides a simple way for us to

communicate with devices on the Internet without remembering complex numbers.

* Every computer on Internet has its own unique address to communicate with it called an IP address.

* An IP address looks like the following 192.168.10.299, 4 sets of digit ranging from 10-255).

* When you want to visit a website, it's not exactly convenient to remember this complicated set of numbers, and that's what DNS can help.

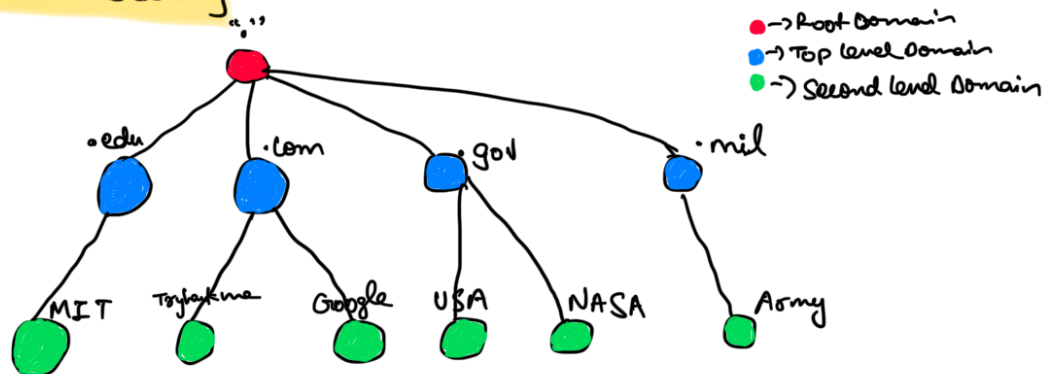
Ex! - Instead of remembering 192.168.10.299, you can remember tryhackme.com instead.

Quiz: -

What does DNS stand for?

A) Domain name System.

Task 2: - Domain Hierarchy



TLD (Top level Domain): - A TLD is the most right hand part of domain name.

Ex! - TryHackme.com TLD is .com. There are 2 types of TLD. 1) generic Top level (gTLD)

2) CCTLD (Country code Top level Domain). 1) Historically a gTLD was meant to tell the user the domain name's purpose. Ex! - .com would be for commercial purpose.

.edu for education purpose and .gov for government purpose.

2) CCTLD was used for geographical purpose Ex:- .ca for Canada site based.
.co.uk for sites based in United Kingdom.

Second level Domain :-

Taking TryHackme.com as an example. The .com part is TLD. And TryHackme is Second level Domain. When registering a domain name, Second level Domain is limited to 63 characters + TLD and can only use a-z-0-9 and hyphen (cannot start or end with hyphen or have consecutive hyphen).

* You can use multiple subdomains split with periods or create longer names. ^{Subdomain (Subdomain and Second level Domain limited to 63 characters)}

Such as jupiter.servers.tryhackme.com. But length must be kept to 253 characters or less. There is no limit number of subdomains you can create for your domain name.

Quiz :-

1) What is the maximum length of Subdomain? | Which character cannot be used in Subdomain (3b-)?
A) 63 | A) _ → underscore

2) What is the maximum length of domain name? | What type of TLD is .co.uk?
A) 253 | A) CCTLD (Country Code Top level Domain)

DNS Record Type - Task 3 :-

* DNS isn't just a website though, multiple types of DNS exist. We will go through most common.

A Record :-

These records resolve to IPV4 address, Ex:- 104.26.10.299

AAAA Record :-

These records resolve to IPV6 address Ex:- 2606:4700:20::681a:be5

CNAME Record :-

This record resolves to another domain name. For example TryHackme's Online Domain has subdomain name store.tryhackme.com which has a CNAME record

Shops.shopify.com. Another DNS request would be made to Shops.shopify.com to work out the IP address.

Ex:- Imagine you opened a small boutique online shop.

→ Your store address is Store.tryhackme.com

→ Instead of building your own warehouse, use Shopify to host and run your shop. Shopify warehouse address is Shops.shopify.com.

MX Record (Mail Exchanger record): -

Tells Internet exactly where to send emails meant for specific domain.

These records resolve to address of servers handle the email for domain you are querying. Ex:- MX record response for tryhackme.com would look something like alt1.aspmx.l.google.com. This record also come with priority flag. This tells client in which order to try servers, This is perfect for if main server goes down and email need to be sent backup server.

TXT Record: -

* TXT records are free text fields where any text based data can be stored.

* TXT records have multiple records, but common ones can be list servers that have authority to send an email on behalf of domain (This help battle against spam and spoofed). They can also be used to verify ownership of domain name when signing up for 3rd party services. Few example.

* _acme-challenge.example.com TXT "token_value_here"

* @TXT "v=spf1 9Pv4:192.0.2.0/24 include:_spf.google.com include:amazon.com null"

you can see as name implies, TXT records are strings of text.

Bug:-

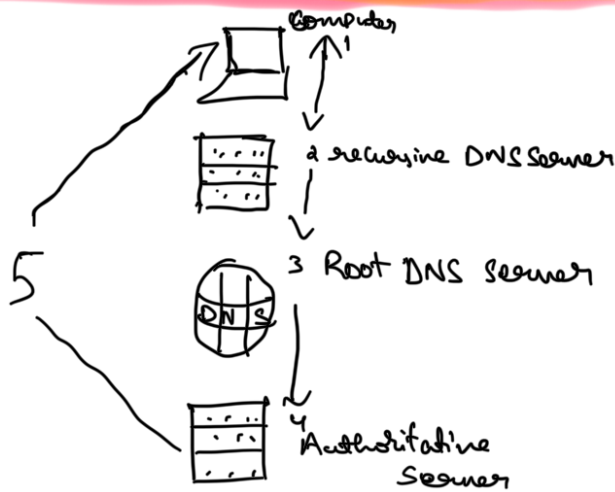
1) What type of record would be advised where to send email? | 2) What type of record handles IPV6 address?

1) MX

2) AAAA

Making a request Task 4 : —

What happens when you make a DNS request : —



- 1) When you request a domain name, your computer first checks its local cache to see if you've previously looked up address recently. If not, a request to recursive DNS server will be made.
- 2) A recursive DNS server is usually provided by your ISP, but you can also choose your own. This server also has local cache of recently looked up domain names. If result is found locally. This is sent back to your computer and your request ends there. If request cannot be found locally, a journey begins to find correct answer, starting with Internet's root DNS servers.
3. Root server act as DNS backbone of Internet, Their job is to redirect you to correct top level domain server, depending on your request. Ex:- request `www.toybankme.com`, root server will recognize Top level domain of `.com` and refer you to correct TLD server that deals with `.com` addresses.
4. The TLD Server holds records to find authoritative servers to answer DNS request. The authoritative server is often also known as name server for domain. Ex:- name server for `Toybankme.com` is `KID.USCloud.com`

and `uma.ns.cloudflare.com`. You'll often find multiple servers for domain name to act as backup in case one goes down.

5. An authoritative DNS server is server that is responsible for storing DNS records for a particular domain name and where any updates to your domain name DNS records would be made. Depending on record type, DNS record is then sent back to recursive DNS server. **DNS records all come with TTL (Time to live) value.**

Quiz:-

1) What field specifies how long a DNS records should be cached for? / 2) What type of server holds all records for a domain?

A) TTL

A) Authoritative

3) What type of DNS server is usually provided by your ISP?

A) recursive

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